

Geography H Review Guide

1 Chapter 2

Demography is the study of the size, structure, growth, distribution, and movement of a population.

Key Issues :

- World's Population Clusters -> major & minor ones
- Two population explosions in human history
- Population Basics
- Demographic Transition Model, Population Pyramid
- Malthus's theory, Population control
- Epidemiological transition

Ecumenes: The ecumene is land that is permanently occupied and used by people for living, working, and economic activities.

Non-Ecumenes: Areas that are not part of the ecumene are typically **too dry, too cold, too high, too hot, or too wet** for permanent human habitation (e.g., deserts, polar regions, high mountains, rainforests).

1.1 Clusters in the World

Cluster Type	Region/Area Included
Major	East Asia (China, Japan (Tokaido Corridor), Korea)
Major	South Asia (India, Pakistan, Bangladesh, Sri Lanka)
Major	Europe (UK to western Russia)
Major	North America (Boswash Corridor, Southeastern Canada)
Minor	Southeast Asia (Indonesia, Philippines, Vietnam, Thailand)
Minor	West Africa (Nigeria and surrounding countries)

Major Clusters have high urbanization rate.

1.2 Different Density Types

Density Type	Definition	Calculation Formula	What It Shows
Arithmetic Density	Total number of people per unit area of land	$\text{Population} \div \text{Total land area}$	Overall population density of a country or region
Physiological Density	Number of people per unit area of arable (farmable) land	$\text{Population} \div \text{Arable land area}$	Pressure on productive land; potential for food shortages
Agricultural Density	Number of farmers per unit area of arable (farmable) land	$\text{Number of farmers} \div \text{Arable land area}$	Efficiency of a region's agriculture; development level

1.3 The Two Population Explosions

Population Explosion	Time Period	Main Causes & Features	Impact on Population Growth
Industrial Revolution	Late 18th–19th century	Advances in industry, medicine, public health, and agriculture; decline in death rates, urbanization, improved food supply	Rapid population growth in Europe and beyond; population doubled and then tripled
Green Revolution	1940s–1970s	Introduction of high-yield crop varieties, chemical fertilizers, pesticides, irrigation, and mechanization	Massive increase in food production; supported rapid population growth, especially in developing countries

1.4 Population Basics

Indicator	Abbreviation	Definition	Formula/Unit	What It Shows	Key Factors Affecting Change
Crude Birth Rate	CBR	Live births per 1,000 people per year	$(\text{Births} \div \text{Population}) \times 1,000$	General fertility	Increase: Improved health, youthful population, pro-natalist cultures. Decrease: Economic development, female education, contraception access, urbanization, high cost of raising children.
Total Fertility Rate	TFR	Average children a woman has in her lifetime	$\text{Sum of ASFRs} \times 5$	Average family size, replacement	Increase: Strong pro-natalist views, limited female opportunities, early marriage. Decrease: Female empowerment, contraception, economic pressures, changing norms.
Age-Specific Fertility Rate	ASFR	Live births per 1,000 women in a specific age group per year	$(\text{Births to women in age group} \div \text{Number of women in age group}) \times 1,000$	Fertility patterns by age	Age at marriage, education, employment, family planning access, health within age group.
Replacement Level	—	TFR needed for population stability (no migration)	-	Population stability	Usually around 2.1; influences if a population will grow or decline naturally.
Doubling Time	—	Years for population to double (constant growth)	$70 \div \text{NIR (Rule of 70)}$	Speed of population growth	Shorter: High Natural Increase Rate (NIR). Longer: Low or negative NIR.
Crude Death Rate	CDR	Deaths per 1,000 people per year	$(\text{Deaths} \div \text{Population}) \times 1,000$	General mortality	Increase: Epidemics, disaster, conflict, poor health access, aging population. Decrease: Medical advances, public health, nutrition, living standards.
Infant Mortality Rate	IMR	Infant (<1 year) deaths per 1,000 live births	$(\text{Infant deaths} \div \text{Live births}) \times 1,000$	Infant health, healthcare quality	Increase: Poor maternal health/care, sanitation, disease, weak healthcare. Decrease: Better prenatal/postnatal care, immunization, nutrition, hygiene, education.
Natural Increase Rate	NIR	Population growth (excluding migration)	$(\text{CBR} - \text{CDR}) \div 10$	Annual % population change	Increase: CBR >> CDR. Decrease: CDR > CBR, or rapidly falling CBR. Often reflects development level.

1.5 Demographic Graph

Demographic momentum (also called population momentum) is the tendency for a population to continue growing for some time after fertility rates have declined to replacement level, due to the existing age structure of the population.

Factor	Effect on Demographic Momentum
Large young population	High demographic momentum, continued growth
Small young population	Low demographic momentum, possible population decline
Declining fertility	Growth slows, but momentum may keep population rising for years
Stable/aging population	Little or no demographic momentum, possible decline

1.6 Maternal Mortality Ratio

Which ratio is the greatest health disparity between LDCs & MDCs? CBR, CDR, Or TFR?

Answer: **Maternal mortality ratio** (*100,000 live births)

1.7 What is the Dependency Ratio?

The dependency ratio measures the proportion of non-working-age people (children and elderly) to working-age people in a population, highlighting the potential economic and social support burden on the productive segment of society.

$$\text{Dependency Ratio} = \frac{\text{Population aged 0-14} + \text{Population aged 65+}}{\text{Population aged 15-64}} \times 100 \quad (1)$$

- **High Dependency Ratio:** Indicates a larger proportion of dependents, meaning more economic pressure on the working-age population to support the young and elderly through taxes, social security, and other means.
 - **Low Dependency Ratio:** Suggests fewer dependents per worker, potentially allowing for greater investment in economic growth, savings, or improved social services.
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1.8 What is the Elderly Dependency Ratio?

The **elderly dependency ratio** (also called the old-age dependency ratio) measures the proportion of elderly people—typically those aged 65 and older—relative to the working-age population, usually defined as those aged 15 to 64. This ratio indicates the potential economic burden on the working-age population to support the elderly, who are generally retired and economically inactive.

$$\text{Elderly Dependency Ratio} = \frac{\text{Population aged 65 and over}}{\text{Population aged 15-64}} \times 100 \quad (2)$$

- **High elderly dependency ratio:** Indicates a greater burden on the working-age population to provide for pensions, healthcare, and other needs of the elderly.
 - **Low elderly dependency ratio:** Suggests a smaller proportion of elderly dependents, meaning less pressure on the working-age population.
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1.9 What is the Youth Dependency Ratio?

The **youth dependency ratio** measures the number of young people (aged 0–14) relative to the working-age population (aged 15–64). It indicates the economic and social burden on the working-age population to support children through education, healthcare, and other services.

$$\text{Youth Dependency Ratio} = \frac{\text{Population Aged 0–14}}{\text{Population Aged 15–64}} \times 100 \tag{3}$$

- Highlights the pressure on resources for childcare, education, and family support systems. A higher ratio means more children depend on each working-age person.

1.10 Four Stages of the Demographic Transition Model (DTM)

The **Demographic Transition Model (DTM)** describes how birth and death rates shift as societies industrialize, leading to population changes. Here are the four stages:

Stage	Name	Birth Rate	Death Rate	Population Growth	Example Countries/Regions
1	High Stationary	High	High	Stable or slow	Pre-industrial societies (e.g., 18th-century Europe)
2	Early Expanding	High	Falling	Rapid increase	Niger, Afghanistan, Sub-Saharan Africa
3	Late Expanding	Falling	Low	Slowing growth	India, Brazil, Mexico
4	Low Stationary	Low	Low	Stable or slow	USA, France, South Korea

1.11 Key Assumptions of the DTM:

1. **Linear progression:** All countries follow the same path from Stage 1 to 4 as they develop economically.
2. **Industrialization-driven:** Declining death/birth rates are tied to technological and social advancements (e.g., healthcare, education).
3. **No migration impact:** The model ignores migration’s role in population changes.

1.12 Stage 5?

Some scholars propose **Stage 5** (post-industrial decline), where:

- **Birth rates** fall below **death rates**, causing a **natural decrease** in population.
- Examples: Japan, Germany, Italy, where death rates exceed births, but total population is often maintained by immigration.
- **Criticism:** The DTM doesn’t account for migration, so Stage 5 remains debated.

1.13 Relating DTM to Population Pyramid

Each DTM stage corresponds to distinct population pyramid shapes:

DTM Stage	Population Pyramid Shape	Characteristics
1	Wide base , narrow top	High youth dependency; short life expectancy
2	Triangular	Youth bulge due to falling death rates
3	Bell-shaped	Narrowing base as birth rates decline
4	Rectangular/columnar	Balanced age groups; aging population
5	Inverted pyramid	More elderly than youth; shrinking workforce

1.14 Malthus's Theory

Thomas Malthus argued in 1798 that:

- Population grows **exponentially** (e.g., 2, 4, 8, 16...).
- Food supply grows **arithmetically** (e.g., 2, 4, 6, 8...).
- This imbalance leads to "positive checks" (famine, disease) and "preventative checks" (late marriage).

Criticism:

- Technological advances (e.g., Green Revolution) disproved Malthus's food scarcity predictions.
- Ignores global trade and family planning's role in population control.

1.15 Four Stages

Stage	Name/Description	Crude Birth Rate (CBR)	Crude Death Rate (CDR)	Natural Increase Rate (NIR)	Causes/Characteristics	Example Countries/Regions
1	High Fluctuating Low Growth	High (40–50 / 1,000)	High (40–50 / 1,000)	None (0 / 1,000)	Lack of family planning, high infant mortality, need for farm workers, disease, famine, war	Britain prior to 1750s
2	Early Expanding High Growth	High (40+ / 1,000)	Lower (15 / 1,000)	High (25+ / 1,000)	Improved sanitation, food production/storage, medical advances, lower infant mortality	Britain 1750s–late 19th c.; Bangladesh, Nigeria today
3	Late Expanding Decreasing Growth	Lower (20 / 1,000)	Low (10 / 1,000)	Low (10 / 1,000)	Higher living standards, better healthcare, education, family planning, mechanization, pensions	Britain late 19th c.; China, Brazil today
4	Low Fluctuating Low Growth	Low (10 / 1,000)	Low (10 / 1,000)	None (0 / 1,000)	Settled economy, developed middle class, political stability, zero population growth	USA, Sweden, Japan, Britain today

2 Chapter 3

2.1 Ravenstein's Laws

Law No.	Law/Principle	Explanation
1	Most migration is over a short distance.	Migrants tend to move short distances, often within their own country, rather than across great distances.
2	Migration occurs in steps.	Migration is not usually a single, direct move to a far-off destination. Instead, people often move in a series of shorter steps, gradually getting closer to their ultimate goal (e.g., from a village to a town, then to a city).
3	Long-range migrants usually move to urban areas.	When people do migrate long distances, they tend to head towards major centers of commerce and industry (cities) where economic opportunities are perceived to be greater.
4	Each migration flow produces a compensating counter-flow.	For every stream of migration in one direction, there is usually a smaller, reverse flow of migrants moving back or in the opposite direction (e.g., return migration, people leaving cities).
5	Rural dwellers are more migratory than urban dwellers.	People living in rural areas are generally more likely to migrate than those already living in urban areas, often seeking better opportunities in cities.
6	Within their own country, females are more migratory than males, but males are more migratory over long distances.	Women tend to migrate more within their own country (often for reasons like family, marriage, or domestic work), while men are more likely to embark on long-distance or international migrations.
7	Most migrants are adults.	The majority of migrants are young adults, often seeking employment or educational opportunities.
8	Large towns grow more by migration than by natural increase.	Major urban centers experience significant population growth due to the influx of migrants rather than solely from births exceeding deaths.
9	Migration increases with economic development.	As a country develops economically, internal migration (and often international migration) tends to increase as people seek out new opportunities.
10	Migration is mostly due to economic causes.	The primary driver for most migration is economic factors, such as the search for better jobs, higher wages, or improved living conditions.

2.2 Real-World Migration

Migration Type	Definition	Key Drivers/Reasons
International	Movement across national borders.	Conflict, persecution, economic opportunity, better living standards, political stability, regional agreements.
Interregional	Movement from one major region of a country to another major region within the same country.	Economic opportunities, climate, lower cost of living, government policy, industrialization.
Intraregional	Movement within a single region of a country (e.g., within a metropolitan area, or within a specific state/province).	Desire for larger homes, better schools, lower crime, quieter lifestyle, remote work, changing life stages, housing affordability.

Chain migration is when immigrants follow others from their home area to a new place, often based on family or social connections. This can lead to communities of people from the same origin forming in the new location.

Medical brain drain is the emigration of trained healthcare professionals from lower-income countries to higher-income countries. This movement depletes the healthcare workforce in the origin countries, weakening their health systems and worsening health outcomes.

Intervening Obstacle: A barrier that hinders or stops a migrant from reaching their intended destination.

- **Example:** A desert, a mountain range, strict visa requirements, or a lack of funds.

Intervening Opportunity: An alternative attractive destination that arises between a migrant's origin and their intended final destination, causing them to settle there instead.

- **Example:** A migrant aiming for Europe finds a good job and a safe community in a transit country like Turkey, deciding to stay there.
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2.3 ESPN

Factor Category	Description	Push Factors (Reasons to Leave)	Pull Factors (Reasons to Come)
E - Economic	Money, jobs, financial opportunities, standard of living.	High unemployment, low wages, poverty, lack of economic opportunity, high cost of living.	Higher wages, better job prospects, economic growth, lower cost of living, educational/training opportunities.
S - Social	Culture, family, community, discrimination, social services, quality of life.	Discrimination, persecution (religious/ethnic), lack of social services (healthcare, education), family separation, limited social freedoms.	Presence of family/friends (chain migration), cultural affinity, better access to social services, greater social freedoms/acceptance, better quality of life.
P - Political	Government, stability, conflict, human rights, safety.	War, political instability, persecution by government, lack of human rights, corruption, forced displacement, lack of safety/security.	Political stability, safety and security, protection of human rights, good governance, asylum opportunities, freedom from conflict.
N - Environmental	Natural world, climate, resources, disasters.	Natural disasters (floods, droughts, earthquakes), climate change impacts (sea-level rise, desertification), resource scarcity (water, arable land), pollution, environmental degradation making an area uninhabitable.	Safer environment, more fertile land, reliable water sources, less exposure to natural hazards, cleaner environment.

Here's a shortened explanation of those terms:

2.4 Complementarity

Complementarity means one place has something another place needs. For migration, it's often about one area having a surplus of people or skills while another has a demand for them.

2.5 Intervening Opportunity

An **intervening opportunity** is an attractive place along a migrant's intended route that causes them to stop there instead of reaching their original destination.

2.6 Transferability

Transferability refers to how easy or difficult it is to move people, goods, or information between two places. It considers the costs and effort involved in making the move.